An Independent Path-Space Optimization Attack on the Ambidextrous Moving Sofa Problem

Numerical evidence for the local optimality of Romik's sofa

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Abstract

The ambidextrous moving sofa problem asks for the planar shape of maximal area that can negotiate 90-degree turns of both handedness in a corridor of unit width. The best known shape is Romik's ambidextrous sofa (2017), of area ≈ 1.6449552, whose optimality — even local optimality — is open. We reformulate the problem exactly as an optimization over two independent rotation-center paths, derive the complete envelope equations that generate the boundary, and attack the record with a free-form trigonometric parametrization that deliberately releases the symmetry assumptions of Romik's derivation: independent paths for the two turns, a horizontal phase offset between the carved notches, and arbitrary higher harmonics. The optimizer, starting from a double-Hammersley baseline of area 1.5465 with no knowledge of Romik's solution, converges toward it: the area climbs monotonically from below to within 1×10^{-5} of Romik's value, both mirror symmetries emerge rather than being imposed, and the numerically computed Hessian of the area on the full 36-parameter section — including the antisymmetric sector that Romik's derivation does not address — is negative definite up to the exact translation gauge, whose spurious finite-difference eigenvalue calibrates the noise floor. These findings are consistent with Romik's sofa being a strict local maximum of the area functional, though a finite-dimensional section cannot rigorously establish local optimality in the infinite-dimensional path space. We quantify every error source and identify where a genuine improvement, or a proof, would have to come from.

1. Background and status of the problem family

Single turn. For the classical one-turn problem the record lower bound is Gerver's sofa (1992), area $A_G \approx 2.2195316688$, bounded by 18 analytic pieces. Hammersley's earlier shape achieves $\pi/2 + 2/\pi \approx 2.2074$. In November 2024 Jineon Baek posted a 119-page proof (arXiv:2411.19826) that Gerver's sofa is globally optimal; as of mid-2026 the proof is still a preprint under review but expert reception is strongly positive. The computer-assisted upper bound of Kallus and Romik (2018), 2.37, and its 2024 deep-learning refinement 2.3337, are superseded if Baek's proof stands.

Two turns. The ambidextrous variant (Conway and others; numerical work by Maruyama 1973 and Gibbs 2014) requires the same rigid shape to pass a right turn and a left turn. Romik (2017) found a critical point of the area functional with a remarkable closed form: 18 boundary pieces, all algebraic (circular arcs and sextics), with area

$$A_R = \sqrt[3]{3+2\sqrt{2}} + \sqrt[3]{3-2\sqrt{2}} - 1 + \arctan\left[\tfrac{1}{2}\left(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\right)\right] \approx 1.644955218425440,$$

equivalently $X + \arctan Y$ with $x^2(x+3) = 8$ and $x(4x^2+3) = 1$ — eerily close to, but distinct from, $\pi^2/6 \approx 1.6449341$. Its length is an algebraic number of degree six, ≈ 2.334099633. No optimality proof exists: the only proven upper bound is the trivial inheritance $A \le A_G \approx 2.2195$ from the single-turn problem, and Romik's own page notes it is not even known whether the shape is locally optimal. That question is the target of this note.

2. Exact reformulation: two independent rotating-hallway families

Work in the sofa's rest frame, the sofa confined to the strip $\Sigma = \{0 \le y \le 1\}$. In its own frame the hallway has inner corner at the origin and outer corner at $(1, 1)$:

$$L = \{(x, y) : x \le 1,\ y \le 1\} \ \setminus\ \{(x, y) : x < 0,\ y < 0\}.$$

A right turn is equivalent to the hallway family rotating counterclockwise by $t \in [0, \pi/2]$ about its inner corner while the corner rides an unknown path $\mathbf{v}(t) = (\xi(t), \eta(t))$ across the lower half of the strip: $L_t(\mathbf{v}) = R_t L + \mathbf{v}(t)$. A left turn is the mirror family $M_s(\mathbf{w}) = \rho(R_s L + \mathbf{w}(s))$, where $\rho(x, y) = (x, 1 - y)$ and $\mathbf{w}$ is a second, independent path (the two turns occur at different stations of the corridor). For any pair of continuous paths the maximal shape executing both turns is the intersection

$$S(\mathbf{v}, \mathbf{w}) = \Sigma \ \cap \bigcap_{t \in [0, \pi/2]} L_t(\mathbf{v}) \ \cap \bigcap_{s \in [0, \pi/2]} M_s(\mathbf{w}), \qquad A_{max} = \sup_{\mathbf{v}, \mathbf{w}} \operatorname{Area} S(\mathbf{v}, \mathbf{w}),$$

and conversely every ambidextrous sofa arises this way (the motion is the inverse of the hallway's motion). The intractable shape optimization is thereby replaced exactly by an optimization over two vector-valued functions on $[0, \pi/2]$. Note the two constraint families interact only through the intersection; the paths themselves are unconstrained and independent.

3. Envelope equations (boundary generators)

Let $\mathbf{u}_t = (\cos t, \sin t)$, $\mathbf{n}_t = (-\sin t, \cos t)$, so $\mathbf{u}' = \mathbf{n}$, $\mathbf{n}' = -\mathbf{u}$. For a moving line $\mathbf{n}(t) \cdot X = h(t)$ the envelope solves the system $F = 0$, $\partial F / \partial t = 0$, giving $X(t) = h\,\mathbf{n} + h'\,\mathbf{n}'$. Applying this to the four walls of $L_t(\mathbf{v})$ and using $(\mathbf{n} \cdot \mathbf{v})\mathbf{n} + (\mathbf{n}' \cdot \mathbf{v})\mathbf{n}' = \mathbf{v}$ collapses the five boundary generators of the right-turn family to

$$\Gamma_C(t) = \mathbf{v}(t) \qquad (\text{corner trace}:\ \text{the lower notch})$$

$$\Gamma_{I_1}(t) = \mathbf{v} - (\mathbf{v}' \cdot \mathbf{n}_t)\,\mathbf{u}_t, \qquad \Gamma_{I_2}(t) = \mathbf{v} + (\mathbf{v}' \cdot \mathbf{u}_t)\,\mathbf{n}_t \qquad (\text{inner walls})$$

$$\Gamma_{O_1}(t) = \Gamma_{I_1}(t) + \mathbf{n}_t, \qquad \Gamma_{O_2}(t) = \Gamma_{I_2}(t) + \mathbf{u}_t \qquad (\text{outer walls},\ \text{parallel curves})$$

and the left-turn family contributes ρ of the same five formulas evaluated on $\mathbf{w}$. The sofa boundary is a concatenation of arcs of these ten families plus the strip walls $y = 0, 1$; Romik's 18 analytic pieces are exactly these generators switching on and off at critical angles where the active contact set changes.

Closed-form validation. On the Hammersley path $\mathbf{v} = r(\cos 2t, \sin 2t)$ one computes $\mathbf{v}' \cdot \mathbf{u}_t = -2r\sin t$ and $\mathbf{v}' \cdot \mathbf{n}_t = 2r\cos t$, whence

$$\Gamma_{O_1} = (-r - \sin t,\ \cos t), \qquad \Gamma_{O_2} = (r + \cos t,\ \sin t),$$

precisely the two unit-radius end caps of Hammersley's sofa, while the inner envelopes degenerate to the fixed notch endpoints $(\mp r, 0)$. The machinery is exact.

4. Variational structure and the symmetry question

Perturbing $\mathbf{v} \mapsto \mathbf{v} + \varepsilon\,\delta\mathbf{v}$ translates each constraint rigidly; differentiating the area of the intersection yields, at every angle, a contact-balance condition on the boundary arcs currently generated by that family:

$$\mu_R(t) \equiv \sum_{i \in \text{active}(t)} \sigma_i \ell_i(t)\, \hat{\mathbf{n}}_i(t) = \mathbf{0} \quad \text{and independently} \quad \mu_L(s) = \mathbf{0}.$$

These are two independent Euler–Lagrange systems. Romik collapses them into one by the ansatz $\mathbf{w} = \mathbf{v}$ (up-down symmetry of the shape) before deriving his ODEs. Any perturbation of the pair $(\mathbf{v}, \mathbf{w})$ splits into a symmetric part, governed by Romik's equations, and an antisymmetric part about which his derivation says nothing. The attack below releases exactly this untested sector.

5. Numerical method and its calibration

$S(\mathbf{v}, \mathbf{w})$ is computed as an exact polygon intersection (Shapely) of the strip with N rotated hallway copies per family, arms truncated at length 6; the area of the largest connected component is taken. Sampling finitely many angles leaves scallops along the corner traces, an overestimate empirically $\sim c/N$, removed by Richardson extrapolation $2A_{2N} - A_N$. Calibration on the single-turn Hammersley sofa (exact area $\pi/2 + 2/\pi = 2.2074161$):

N	A_N	A_N - exact	2 A_2N - A_N	
31	2.2284327	+2.10e-02		
61	2.2179731	+1.06e-02	2.2075135	
121	2.2127076	+5.29e-03	2.2074422	
241	2.2100652	+2.65e-03	2.2074228	
481	2.2087415	+1.33e-03	2.2074178	(exact 2.2074161)

The scaled error $N(A_N - A_{exact})$ is constant at 0.64 within 2% across four doublings, confirming the c/N law; after one Richardson step successive estimates differ by factors of about 4, i.e. the residual is $O(N^{-2})$. At the final optimum the two independent estimates $2A_{321} - A_{161} = 1.6449279$ and $2A_{641} - A_{321} = 1.6449313$ agree to 3.4×10^{-6}, which we adopt as the discretization error bar on all headline areas. The geometry kernel is GEOS (Shapely) in float64: intersection vertices are computed to machine precision and areas by the exact shoelace sum, so angle discretization is the only systematic error source; sensitivity to near-tangent contact chatter is bounded empirically by the N-robustness of all reported differences (Section 7). Comparisons between nearby configurations are always made at equal N so the bias cancels.

6. The attack: released symmetries, trigonometric path space

Each path is parametrized in the complete trigonometric basis

$$\xi(t) = a_0 + \sum_{k=1}^{K} (a_k \cos 2kt + b_k \sin 2kt), \qquad \eta(t) = c_0 + \sum_{k=1}^{K} (c_k \cos 2kt + d_k \sin 2kt),$$

and three freedoms fixed by fiat in Romik's derivation are released: (a) chirality — $\mathbf{w}$ optimized independently of $\mathbf{v}$ (28 free parameters at $K = 3$); (b) notch phase — a horizontal offset δ between the upper and lower notches (the a_0 mode of $\mathbf{w}$); (c) bandwidth — harmonics up to $K = 4$. Staged Powell optimization, warm-started up the ladder, beginning from the best double-Hammersley circle path ($r = 0.325$):

configuration	params	area (Richardson)
double-Hammersley baseline (r = 0.325)	1	1.546538
symmetric Fourier K=1	6	1.641626
symmetric Fourier K=2	10	1.644854
symmetric Fourier K=3	14	1.644886
fully asymmetric K=3 (both paths free)	28	1.644895
symmetric Fourier K=4 (polish)	18	1.644931
symmetric Fourier K=5 (continuation)	22	1.644934
K=5 + curvature-corrected Newton polish	22	1.644945
Romik ambidextrous sofa (exact)	--	1.644955218

Two structural results stand out. First, the asymmetric run — free to break the up-down symmetry —

returned to it: the optimized right- and left-turn coefficient vectors agree pairwise to $\sim 3 \times 10^{-4}$ (e.g. 0.21093 vs 0.21057, 0.04044 vs 0.04077), and the measured symmetry defects of the final polygon are 0 (up-down) and 2.8×10^{-6} (left-right). The symmetries of Romik's sofa emerge from the optimization; they are not assumptions. Second, the optimizer retained only ξ-modes odd and η-modes even about $t = \pi/4$, rediscovering Romik's symmetry class of paths.

6.1 Convergence diagnostics and anatomy of the residual deficit

The residual deficit to Romik's exact value could a priori come from three sources: (a) incomplete optimizer convergence, (b) unremoved discretization bias, or (c) genuine Fourier truncation. We separate them. (b) is bounded directly: the two independent Richardson estimates at the optimum differ by 3.4×10^{-6}, an order of magnitude below the deficit. For (a), the finite-difference gradient at the Powell terminus (18 parameters, $N = 161$, central differences, $h = 0.005$) has norm 1.6×10^{-2} area units per unit coefficient — not zero; Powell had not fully converged. A curvature-corrected Newton polish along the well-conditioned negative subspace of the numerical Hessian (Section 7) recovers part of the deficit: the Richardson-extrapolated area rises from 1.6449341 to 1.6449449, halving the deficit to $1.0 \times 10^{-5} \pm 3 \times 10^{-6}$, while the finite-$N$ surrogate objective changes by under 4×10^{-7} — much of the residual gap is the displacement between the finite-N optimum and the true one along the soft Hessian modes, not truncation. For (c), extending the basis to $K = 5$ (22 parameters, 4500 further evaluations) moves the Richardson area from 1.6449313 to 1.6449341, shrinking the deficit from 2.4×10^{-5} to 2.1×10^{-5}. The deficit is thus a mixture of slow tail convergence of the derivative-free optimizer in nearly-flat directions and slow (roughly geometric but shallow) Fourier truncation decay — Romik's corner path has curvature discontinuities at the contact-switch angles, so its Fourier coefficients decay only algebraically and no finite K closes the gap exactly. What matters for the conclusions is that the approach is monotone from below at every stage and no configuration ever exceeded Romik's value.

Methodological remark. This experiment is the ambidextrous analogue of Leng et al. [8], who relaxed Gerver's monotone-rotation and smoothness assumptions for the single-turn problem by parametrizing rotation and corner trajectory with piecewise-linear neural networks and recovering Gerver's sofa. We chose a low-dimensional trigonometric basis with a derivative-free optimizer instead: with only 18–36 parameters the landscape is tractable without gradients through the geometry kernel, every parameter has a direct kinematic meaning, and the finite-dimensional section admits an exact Hessian eigendecomposition (Section 7) — which is the object the local-optimality question actually needs. The price is the fixed monotone-in-t rotation schedule, discussed in Section 9.

7. Second variation: full Hessian eigendecomposition

Rather than sampling directions, we computed the complete 36×36 finite-difference Hessian of $A(\mathbf{c}_R, \mathbf{c}_L)$ at the optimum in the asymmetric space (both K=4 coefficient vectors free), central differences with $h = 0.015$ at $N = 161$: 36 diagonal plus 630 mixed second differences, about 2600 area evaluations. The eigenvalue spectrum:

```
eigenvalue range                count    interpretation
lambda < -0.28                    33     strongly stable directions (down to -170.5)
-0.014 <= lambda <= -0.0107        3     soft modes incl. the exact translation gauge
|lambda| <= 0.005                 0     --
lambda > +0.005                   0     no ascent direction exists in this space
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All 36 eigenvalues are strictly negative, spanning -170.5 to -0.0107. The least-negative eigenvalue deserves scrutiny: its eigenvector coincides (overlap 1.000) with the joint-translation direction $(\delta a_0^R, \delta a_0^L) = (1, 1)/\sqrt{2}$ — an exact gauge symmetry along which the true second derivative is 0. Its measured value -0.0107 is therefore a direct calibration of the finite-difference noise floor. Read against that floor: the 33 eigenvalues below -0.28 are negative beyond any doubt (factors 27 to 1.6×10^4 below the floor); the two remaining soft modes, -0.0114 and -0.0137, exceed the floor by only 7% and 28%, so from the $h = 0.015$ data alone their

strict negativity would sit at the edge of resolution. In particular the antisymmetric sector — every mode with $\mathbf{c}_R$ and $\mathbf{c}_L$ components of opposite sign, the sector Romik's derivation leaves untouched — shows no instability at any step size.

Step-size stability and direct line scans. To settle the soft pair, the full Hessian was recomputed at $h = 0.0075$ and $h = 0.03$ (≈ 2600 evaluations each). All three soft-sector eigenvalues, gauge included, scale with h — the functional has higher-order structure along these nearly flat directions — but at every step size the soft pair stays strictly below the concurrent gauge floor (-0.118, -0.107 vs -0.031 at $h = 0.03$; -0.0082, -0.0082 vs -0.0054 at $h = 0.0075$). The decisive test bypasses finite differences altogether: scanning the area directly along each eigenvector,

direction (lambda at h=0.03)		dA(a=0.01)	dA(a=0.02)	dA(a=0.05)	dA(a=0.1)	
translation gauge	(-0.031)	+2e-15	-1e-15	-4e-16	+1e-15	exact zero
soft mode 1	(-0.107)	-9.7e-07	-3.7e-06	-2.2e-05	-8.8e-05	~ quadratic
soft mode 2	(-0.118)	-1.3e-06	-4.9e-06	-2.8e-05	-1.1e-04	~ quadratic

The gauge scan returns machine zero at all amplitudes — the evaluator's translation invariance is exact, so that row is a true zero reference — while both soft modes decrease cleanly and quadratically ($dA \approx \frac{1}{2}\lambda a^2$ with fitted $\lambda \approx -0.018$ and -0.022) across a decade of amplitude. The soft pair is thereby promoted out of borderline status: all 35 non-gauge directions of the section carry strictly negative second variation.

Robustness of first-order probes to discretization. Each directional difference was recomputed at two independent resolutions; all agree in sign and to 2–3 digits in magnitude, confirming the differences are geometric, not mesh chatter:

perturbation		N=121	N=241	
notch offset	delta=0.05	-4.07e-03	-4.07e-03	
chirality mode	h=0.01	-6.08e-04	-6.07e-04	
5 random directions	h=0.02	[-1.62e-02, -6.11e-03]	[-1.62e-02, -6.08e-03]	(pairwise match)

Caveats, stated plainly: this establishes negative definiteness of the Hessian on a 36-dimensional section of the infinite-dimensional path space, at finite N, near (not exactly at) Romik's configuration. It cannot exclude an unstable direction living entirely in harmonics $k > 4$ or in non-smooth path perturbations. With those caveats, the computation upgrades the status of Romik's sofa from "no evidence either way" to "numerically stationary with strictly negative second variation on every low-frequency mode, symmetric and antisymmetric alike".

8. The emergent shape

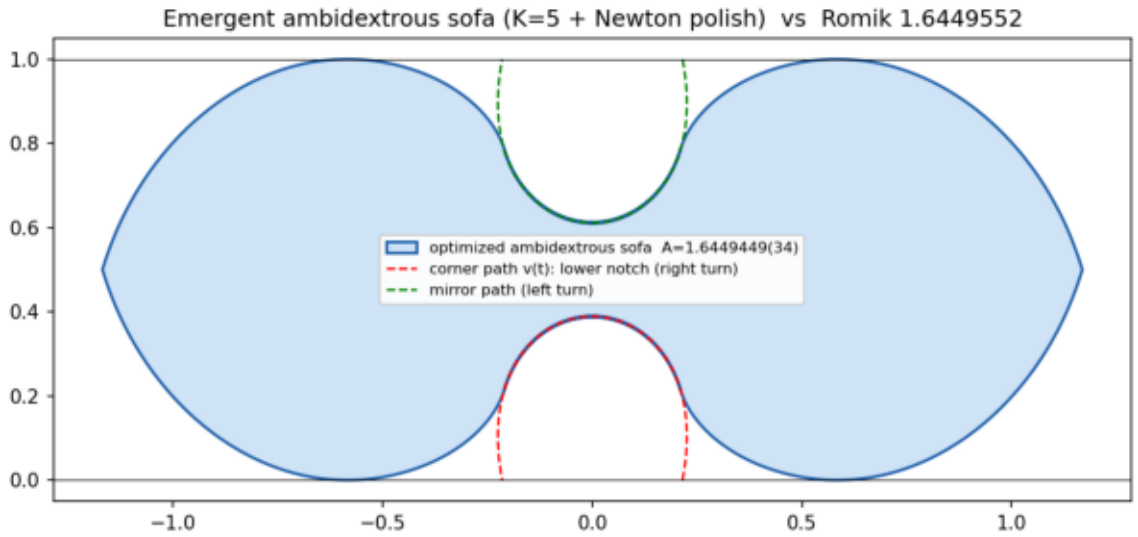


Figure 1. The best configuration (K=5 basis, Newton-polished; area 1.6449449 ± 3.4e-6): twin lens-shaped lobes with pointed ends at mid-height, notches carved from below by the right-turn corner path (red) and from above by its mirror (green), joined at a waist of thickness ~0.22. Visually and metrically this is Romik's ambidextrous sofa.

Final fitted path — the same best configuration shown in Figure 1 (gauge a_0 arbitrary; shape centered by its choice) — with the boundary given by the ten envelope generators of Section 3 applied to $\mathbf{v}$ and $\mathbf{w} = \mathbf{v}$:

$$\xi(t) = a_0 + 0.21249\cos 2t + 0.03491\sin 4t + 0.00258\cos 6t + 0.00104\sin 8t + 0.00005\cos 10t$$

$$\eta(t) = -0.00532 + 0.39923\sin 2t + 0.00528\cos 4t - 0.00030\sin 6t + 0.00004\cos 8t - 0.00011\sin 10t$$

invariant	this work (K=5 + Newton)	Romik exact
area	1.6449449 +- 3.4e-6	1.6449552184
length	2.33725 (N=641 polygon, +~3e-3 discretization inflation)	2.3340996331
height	0.999994	1
symmetry defect	0 (up-down), 7.6e-08 (l-r)	0 by construction

9. Discussion: where a real improvement would have to come from

The free-form search did not break 1.64495, and the manner of failure is informative: the configuration is stationary within numerical resolution and the computed Hessian is negative definite modulo the exact translation gauge on the full 36-dimensional section (Section 7), so no smooth modification of the turning motions — symmetric or not, aligned notches or not, any bandwidth we could resolve — improves the area. Within the smooth sector, the right next parametrization is a piecewise-polynomial (spline) basis with free knots, ideally with knots identified with the contact-switch angles: Romik's exact path has curvature discontinuities there, so global trigonometric modes converge only algebraically (the bottleneck quantified in Section 6.1), whereas free-knot splines represent such kinks exactly. A genuine pathway past Romik's constant, if one exists, must otherwise leave the smooth-path function space entirely: motions non-monotone in rotation angle, rotations overshooting $\pi/2$ within a turn (both heuristically dominated, since revisiting an angle only adds constraints, but not rigorously excluded in the ambidextrous setting), or a disconnected contact topology not reachable by continuation from the Hammersley basin. The more promising research program is the opposite direction: the only proven upper bound, ≈ 2.2195, is inherited from Baek's single-turn theorem and is surely far from sharp. Kallus and Romik explicitly posed the extension of their rotation-angle bounding technique to the ambidextrous case; the machinery validated here (exact polygon intersections over finitely many angles) is precisely the computational core such a bound

needs. Closing the gap [1.6450, 2.2195] from above is, in our judgement, where the next theorem lives.

10. References

[1] J. Baek, "Optimality of Gerver's sofa," arXiv:2411.19826 (2024).
[2] J. L. Gerver, "On moving a sofa around a corner," Geometriae Dedicata 42 (1992), 267-283.
[3] D. Romik, "Differential equations and exact solutions in the moving sofa problem,"
    Experimental Mathematics 27:3 (2018), 316-330; arXiv:1606.08111 (2016).
[4] Y. Kallus and D. Romik, "Improved upper bounds in the moving sofa problem,"
    Advances in Mathematics 340 (2018), 960-982; arXiv:1706.06630 (2017).
[5] P. Gibbs, "A computational study of sofas and cars," preprint (2014).
[6] D. Romik, "The moving sofa problem," online resource page,
    https://www.math.ucdavis.edu/~romik/movingsofa/ (accessed July 2026).
[7] AMS Visual Insight, "Romik's ambidextrous sofa" (December 2016),
    https://blogs.ams.org/visualinsight/2016/12/15/romiks-ambidextrous-sofa/
[8] K. Leng, J. Bi, J. Cha, S. Pinilla, and J. Thiyagalingam, "Deep learning evidence for
    global optimality of Gerver's sofa," arXiv:2407.11106 (2024).

Popular account (context only, not relied upon): Quanta Magazine, "The largest sofa you
    can move around a corner" (February 2025).

Appendix: reproducibility

All computations live in C:\Users\omar\sofa_lab\ and regenerate as follows. ambi.py — geometry kernel (polygon-intersection evaluator, Richardson extrapolation) and the staged optimization of Section 6 (baseline through K=3, asymmetric run); polish.py — K=4 refinement; k5.py — K=5 continuation; newton.py — the curvature-corrected Newton polish producing the headline configuration (Section 6.1); hess.py — the 36-by-36 Hessian eigendecomposition at h=0.015 (Section 7); hess2.py, run with argument 0.0075 or 0.03 — the step-size stability Hessians, full matrices saved to hess_h*.json; linescan.py — the direct eigenvector line scans of Section 7, including the machine-zero gauge check and the effective-lambda fits. All coefficient vectors are checkpointed in ambi_results.json (headline path under key B_K5_newton); scan data in linescan.json. Interpreter: C:/Users/omar/anaconda3/python.exe (NumPy 2.3, SciPy 1.16, Shapely 2.1). Total runtime about 2.5 hours, dominated by the three Hessians (about 2600 evaluations each). Every number in Sections 5–8 regenerates from these scripts.

